

Digital Design with Verilog

Counter Designing using FSM

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Outline

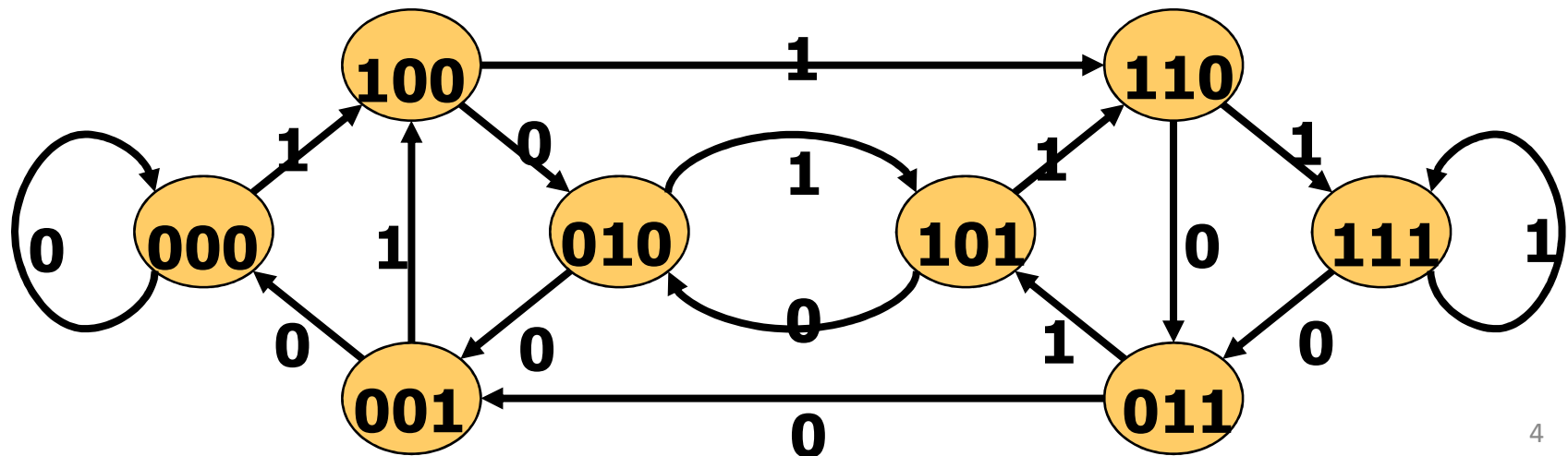
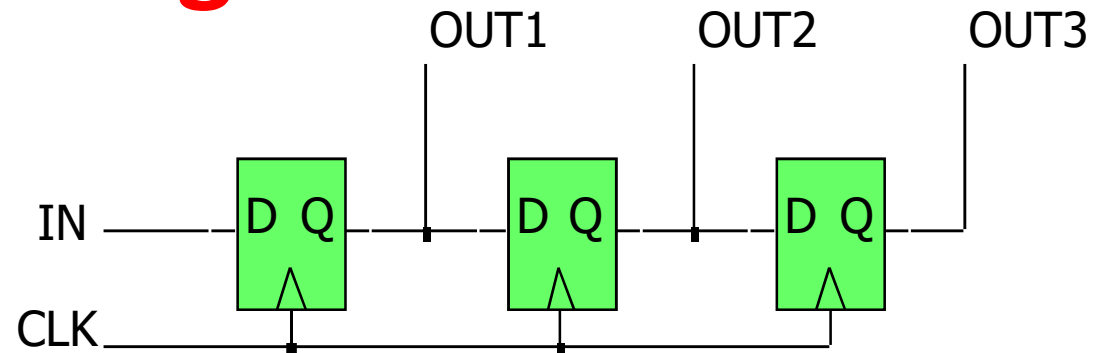
- FSM of Register/Counter
 - **Self Starting Counter**
- FSM Based Counter Design : using diff. FFs
 - **Excitation of FFs : From FSM of FF**
 - Optimizing Circuits: given diff. FFs

**Can any sequential system be
represented with a state diagram?**

YES

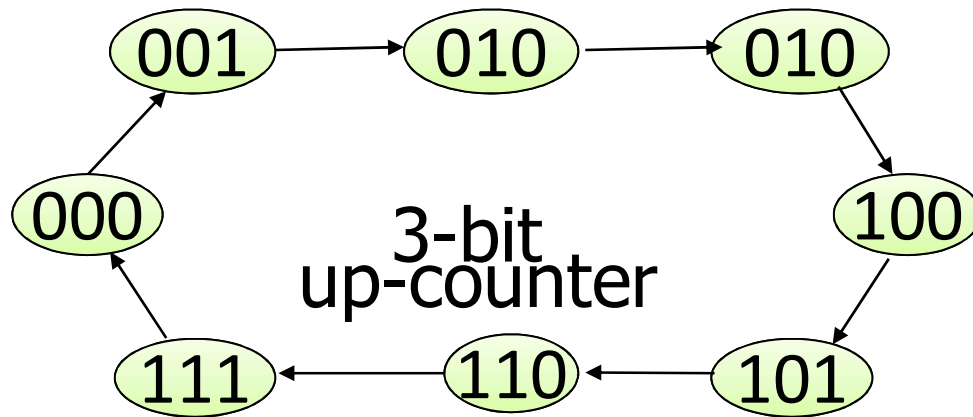
Shift Register

- Input value shown on transition arcs
- Output values shown within state node



FSM for a Counter

- Tabular form of state diagram
- Like truth-table (specify O/P for all input combinations)
- Encoding of states: easy for counters – just use value

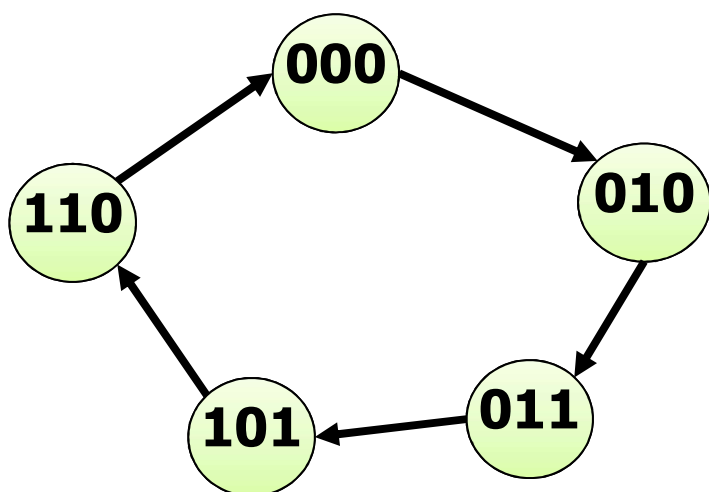


PS	NS
000	001
001	010
010	011
011	100
100	101
101	110
110	111
111	000

Example: 5-state counter

- Counter repeats 5 states in sequence
 - Sequence is 000, 010, 011, 101, 110, 000

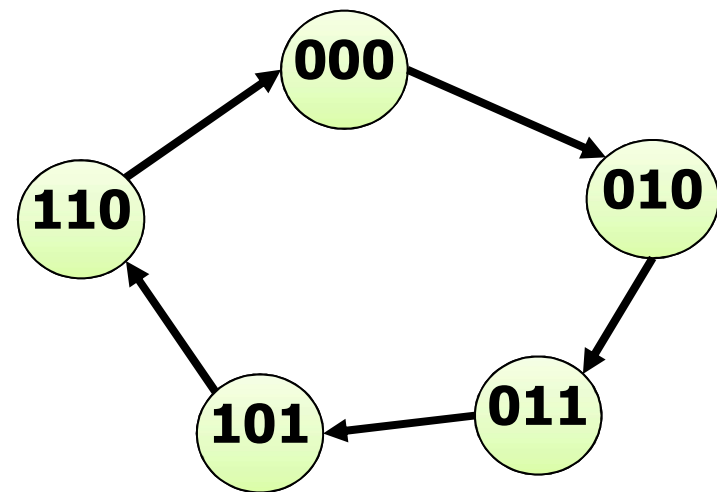
Step 1: State diagram



Step 2: State transition table Assume D flip-flops

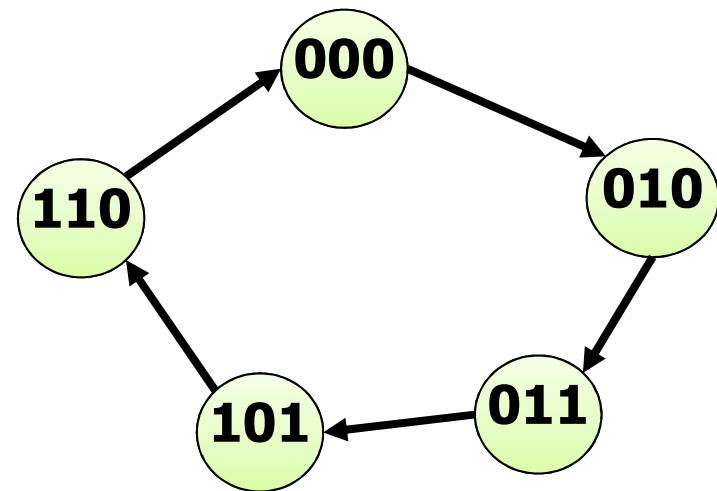
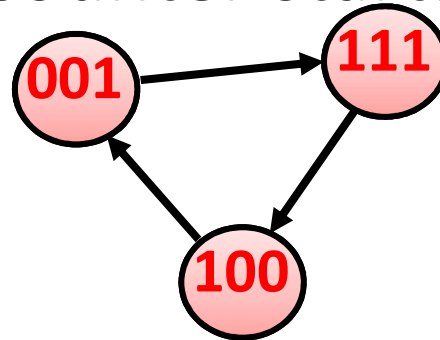
Present State			Next State		
C	B	A	C+	B+	A+
0	0	0	0	1	0
0	0	1	X	X	X
0	1	0	0	1	1
0	1	1	1	0	1
1	0	0	X	X	X
1	0	1	1	1	0
1	1	0	0	0	0
1	1	1	X	X	X

Is our design robust?



Is our design robust?

- What if the counter starts in a 111 state?

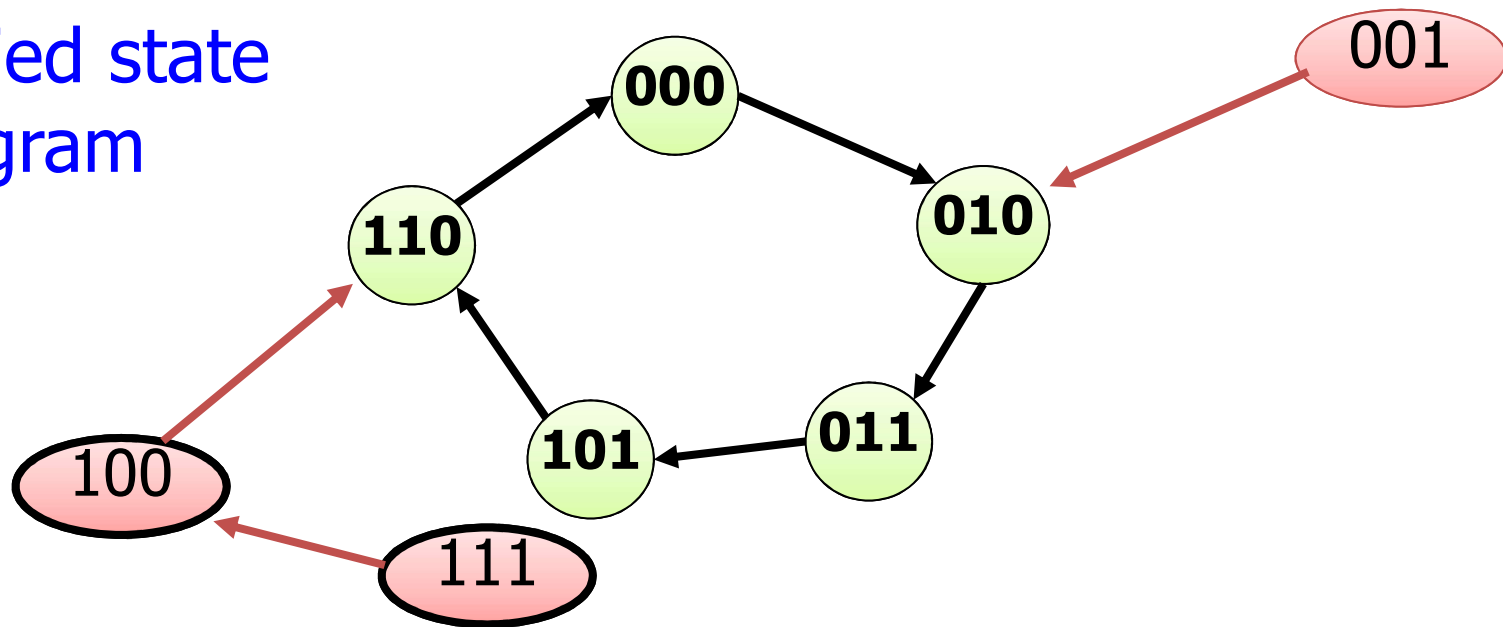


Does our counter get stuck in invalid states???

5-state counter

Proper methodology is to ***design*** your counter to be self-starting

Modified state diagram



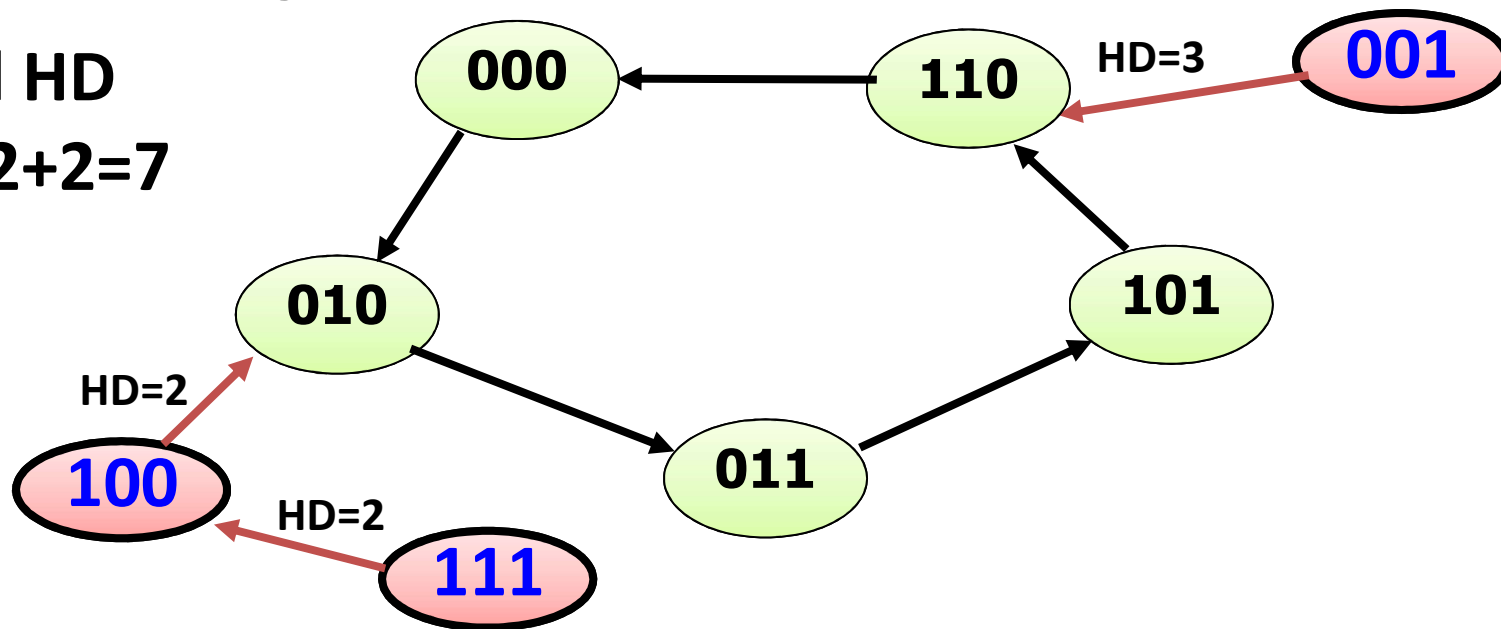
Self-starting counters

- Invalid states should **always** transition to valid states
 - Assures startup,
 - Assures bit-error tolerance
- Design your counters to be self-starting
 - Draw **all** states in the state diagram
 - Fill in the **entire** state-transition table
 - May limit your ability to exploit don't cares
 - Choose startup transitions that minimize the logic : **Hamming Distance Minimization**

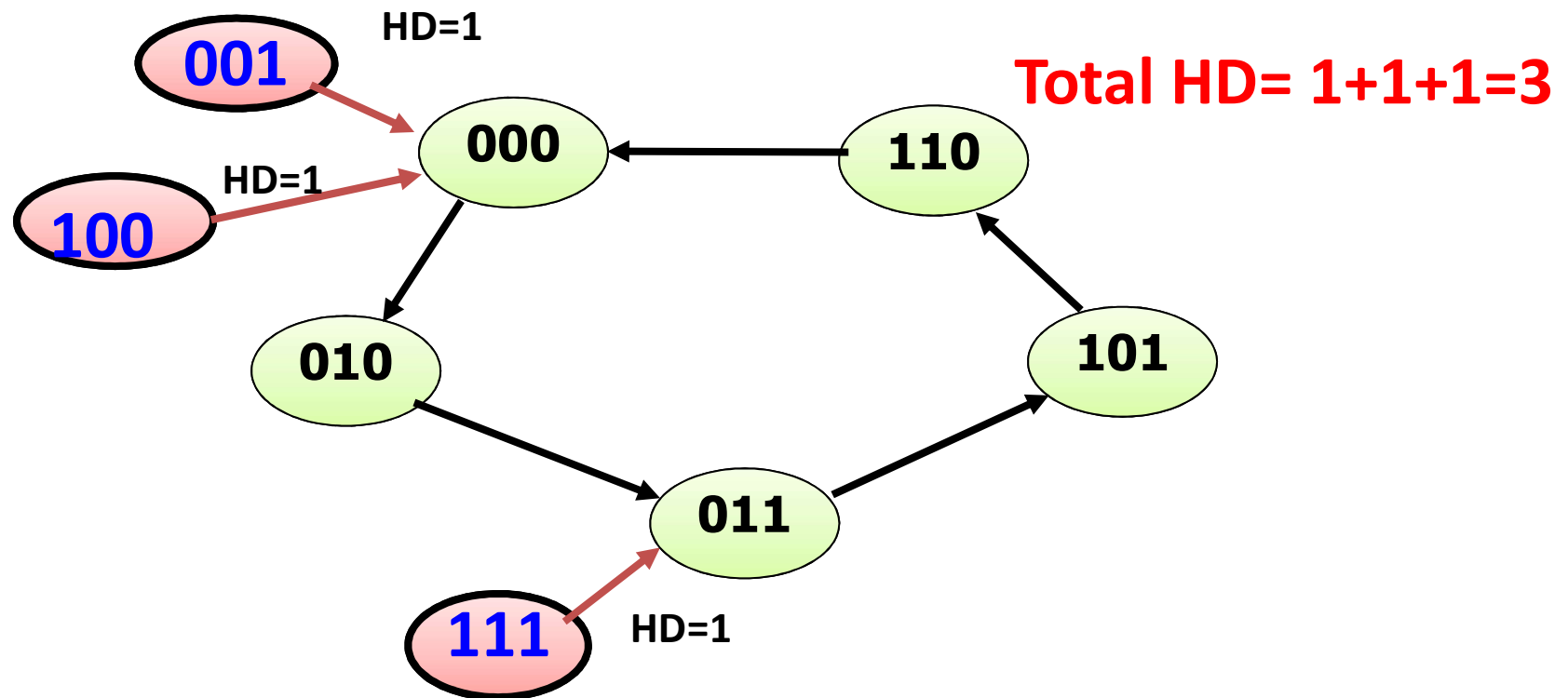
5-state counter

HD=Hamming Distance

Total HD
= 3+2+2=7



5-state counter



Counter design based on FSM using D, T, JK FFs

Reference : Section 6.9 of Book

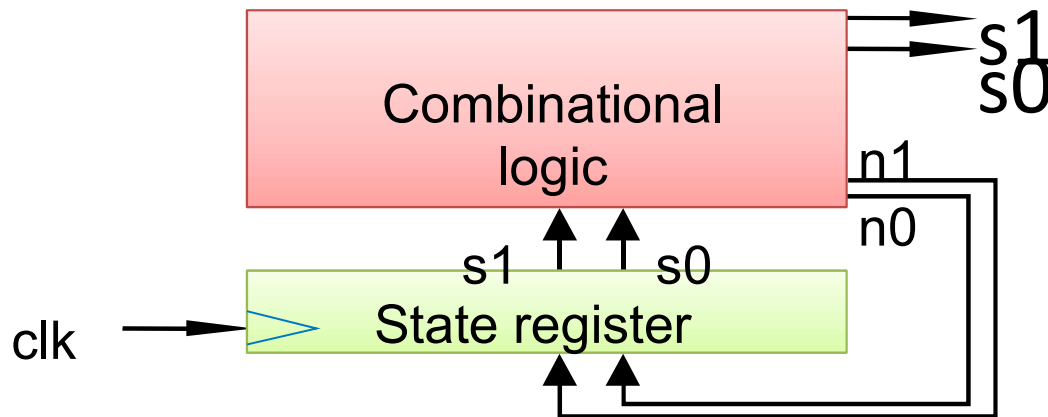
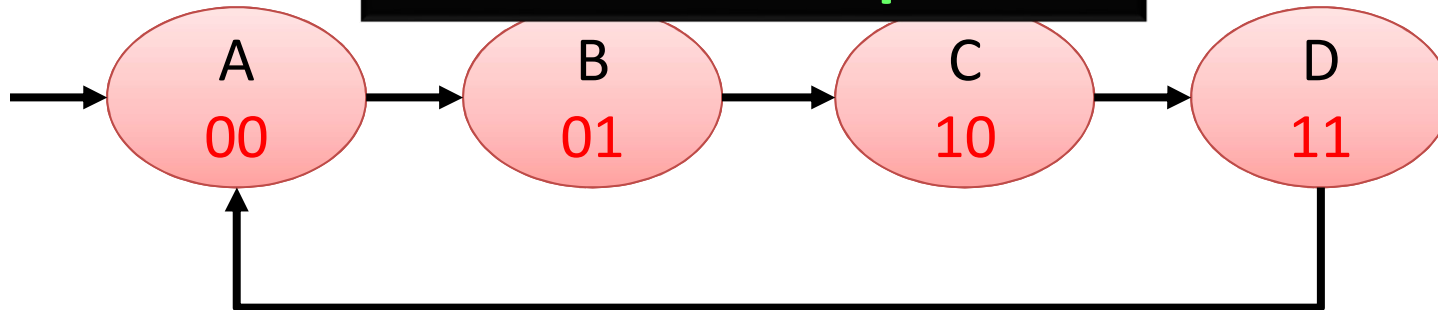
**Donald D. Givone, *Digital Principles and Design*,
McGraw-Hill, 2003**

FSM Implantation

- Default FF is D-FF
 - Simple to design D-FF
- Can be designed using other FF too.
- Benefits : Given other FFs
 - Takes benefit of dual inputs to FF
 - Counter can be implemented using Small Combinational Circuit: Given other FFs
 - More Inputs from Combinational Circuit

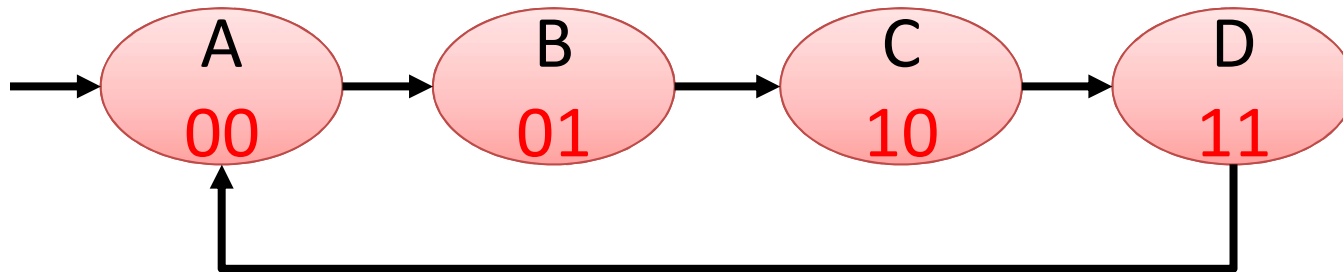
FSM of Counter : 2 bit

State bits = Output bits

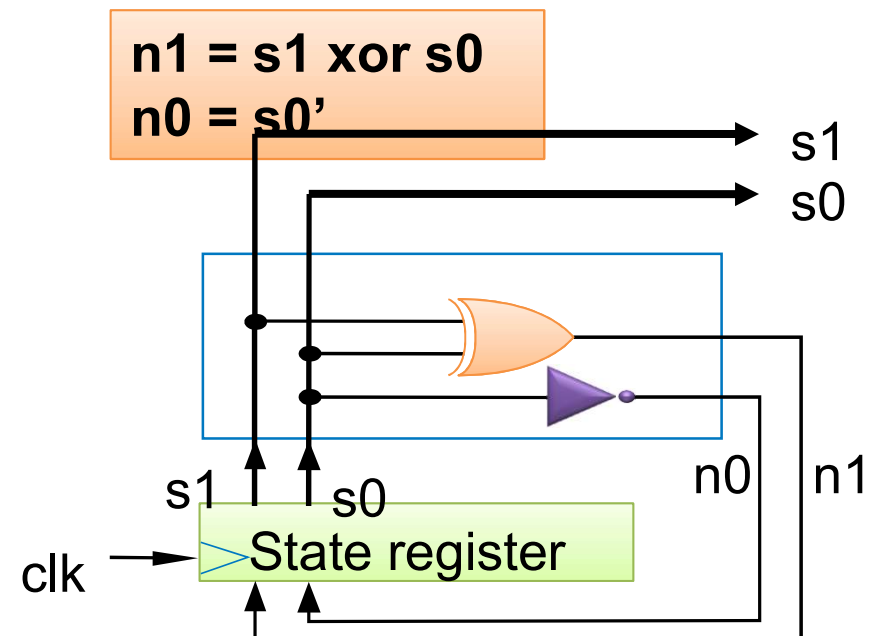


	I/P		O/P	
	s1	s0	n0	n1
A	0	0	0	1
B	0	1	1	0
C	1	0	1	1
D	1	1	0	0

FSM Controller: Binary Counter



	I/P		O/P	
	s1	s0	n0	n1
A	0	0	0	1
B	0	1	1	0
C	1	0	1	1
D	1	1	0	0

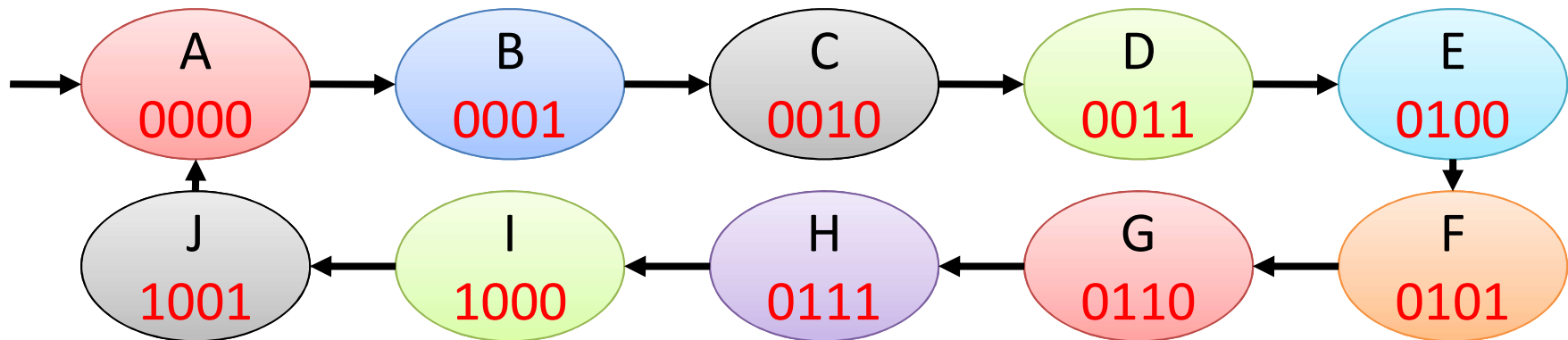


Synchronous Counter : Design

- **Together: Create FSM, Encode Bit**
- State Table
- Design Combination Circuit

Mod 10 Counter: BCD Counter

- Count from 0000 to 1001 (0-9 the Reset)
- FSM with Encoding **done**



State Table Creation

Present State				Next State			
S3	S2	S1	S0	N3	N2	N1	N0
0	0	0	0	0	0	0	1
0	0	0	1	0	0	1	0
0	0	1	0	0	0	1	1
0	0	1	1	0	1	0	0
0	1	0	0	0	1	0	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	1	1
0	1	1	1	1	0	0	0
1	0	0	0	1	0	0	1
1	0	0	1	0	0	0	0

$$N0 = S0'$$

$$N1 = S1S0' + S3'S1'S0$$

$$N3 = \dots \text{Solve}$$

$$N4 = \dots \text{Solve}$$

Using other FF in Counter

- Takes benefit of dual inputs to FF
- Counter can be implemented using **Small Combinational Circuit**: Given other FFs
- More Inputs from Combinational Circuit
- **Use of Excitation Table**
 - How FF out changes from one to others
 - Required FF inputs to change FF output
 - 0 to 0, 0 to 1, 1 to 0 and 1 to 1

Excitation Table for T Flip Flop

- It is different than Characteristic Equation
- Tabling requires **Input** to change from **Q** to **Q⁺**

T	Q ⁺
0	Q _t
1	Q _t '

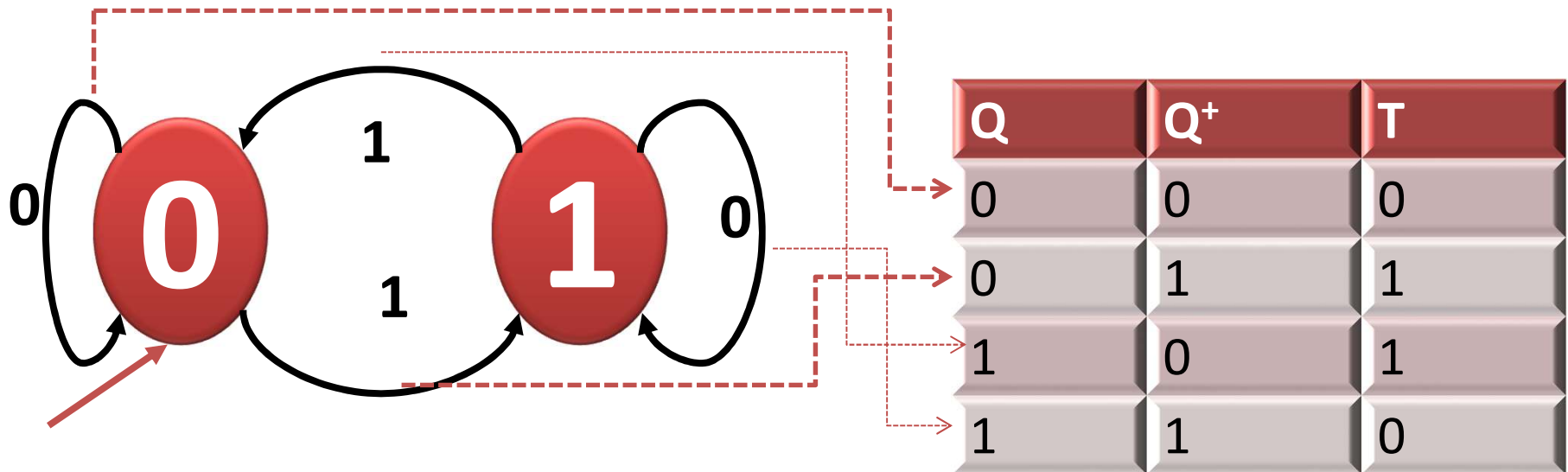
Q	Q ⁺	T
0	0	0
0	1	1
1	0	1
1	1	0

Characteristic Table

Excitation Table

Excitation Table for T Flip Flop

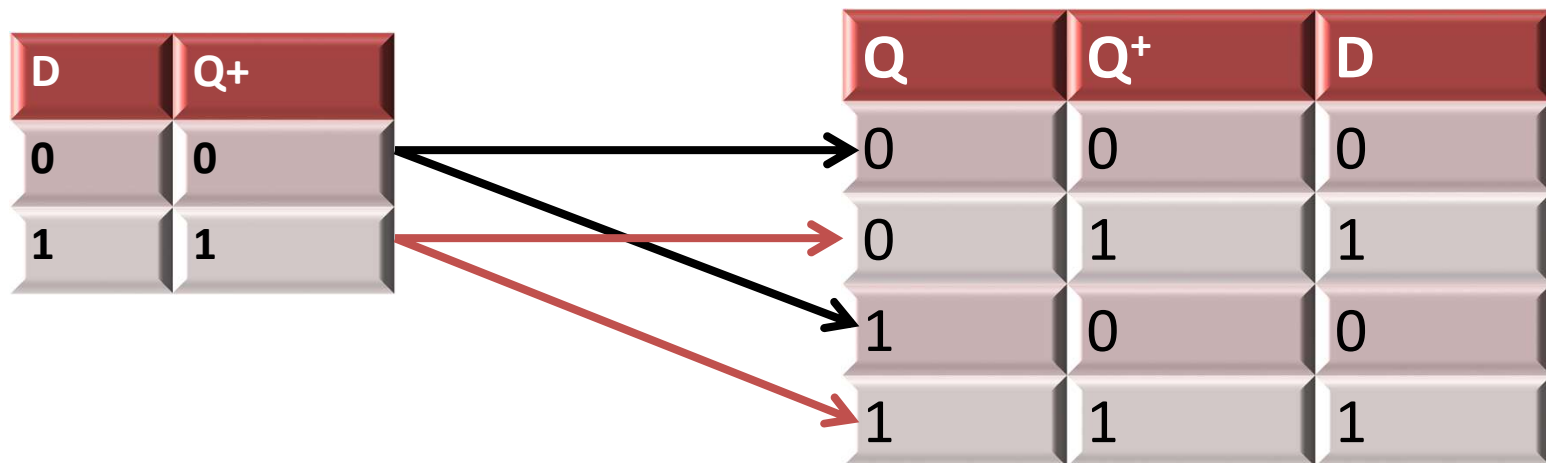
- Tabling requires **Input** to change from **Q** to **Q⁺**



Excitation Table

Excitation Table for D Flip Flop

- Tabling requires **Input** to change from **Q** to **Q⁺**



Characteristic Table

Excitation Table

Excitation Table for JK Flip Flop

- Tabling requires **Input** to change from **Q** to **Q⁺**

J	K	Q ⁺
0	0	Q _t
0	1	0
1	0	1
1	1	Q _t '

Characteristic Table

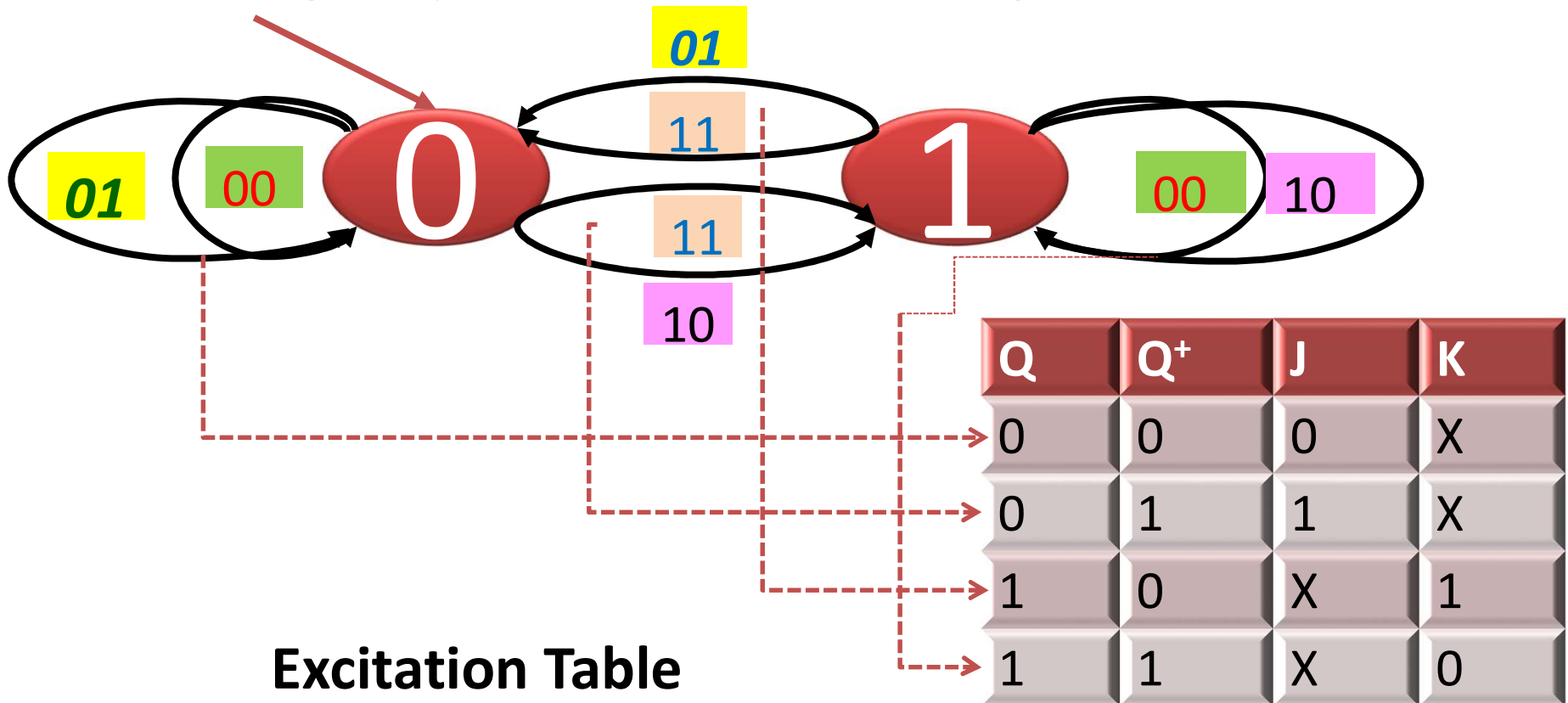


Q	Q ⁺	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Excitation Table

Excitation Table for JK Flip Flop

- Tabling requires **Input** to change from **Q** to **Q⁺**



Excitation Table for SR Flip Flop

- It is different than Characteristic Equation
- Tabling requires **Input** to change from **Q** to **Q⁺**

S	R	Q ⁺
0	0	Q _t
0	1	0
1	0	1
1	1	U

Characteristic Table

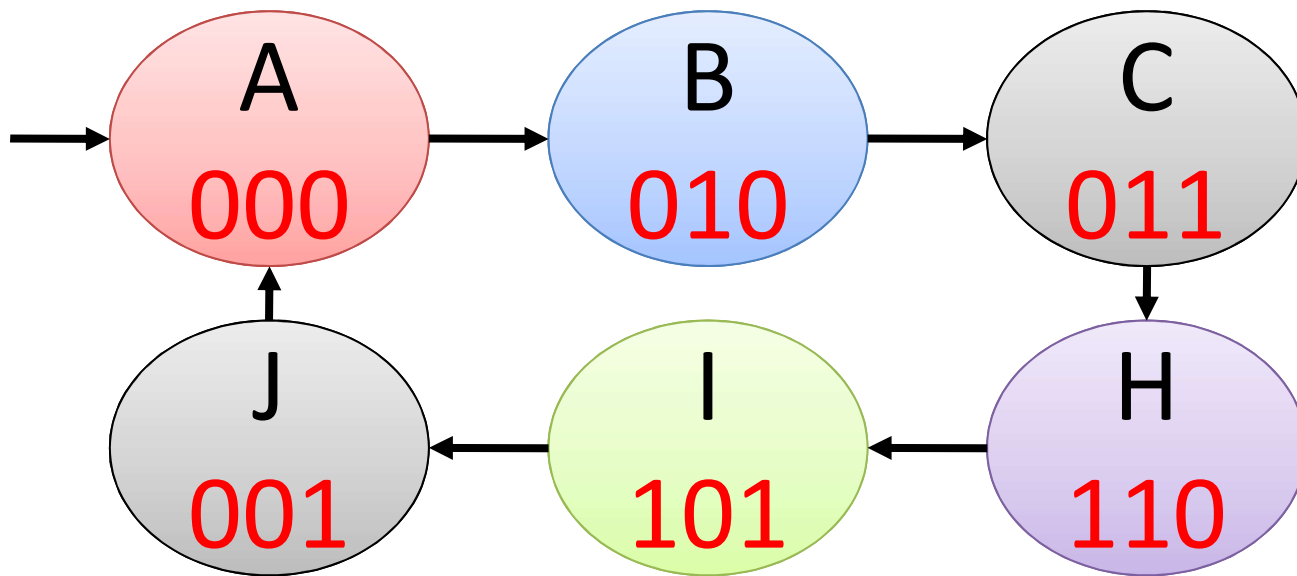


Q	Q ⁺	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

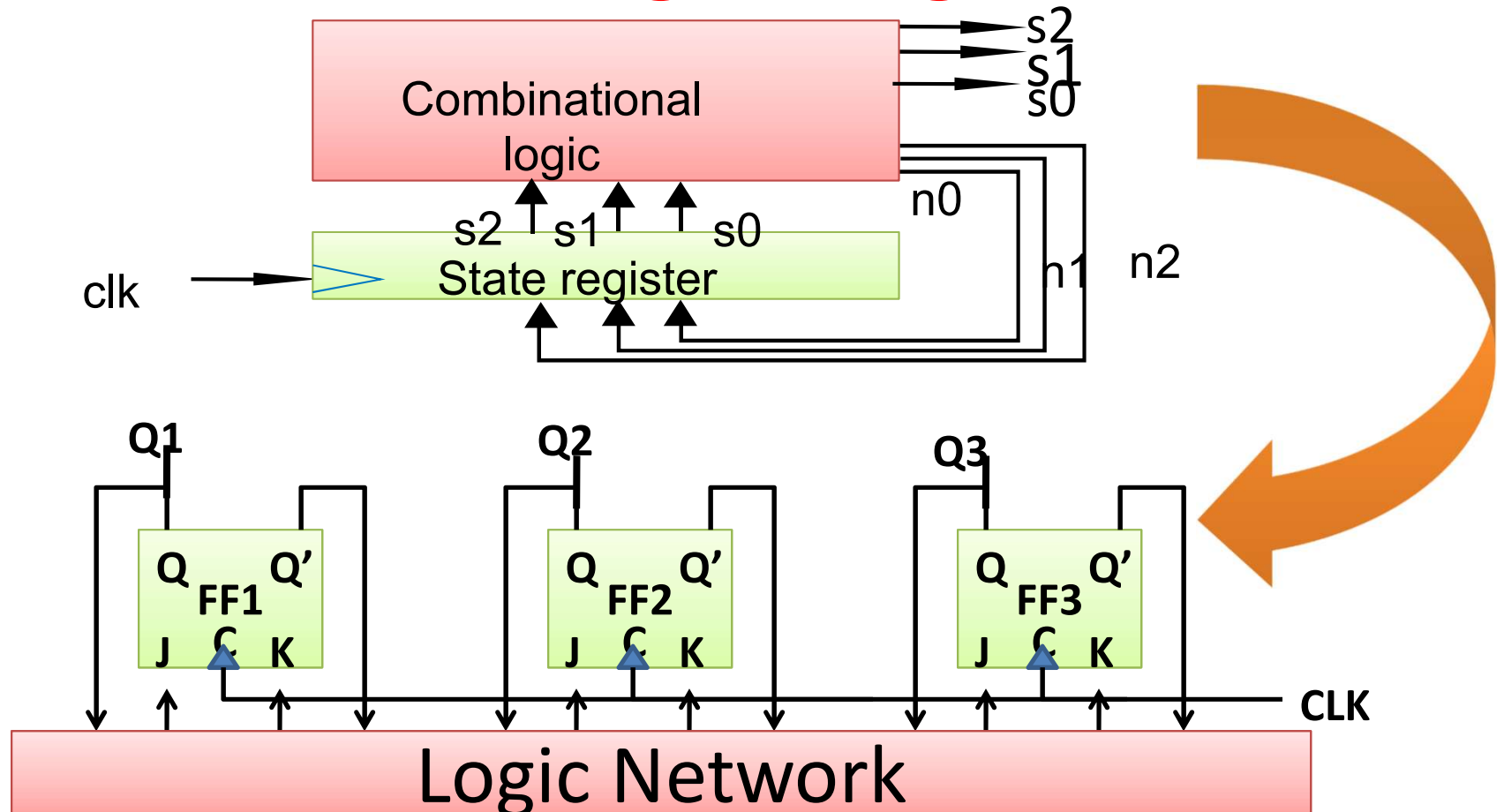
Excitation Table

Design of Counter: With FFs

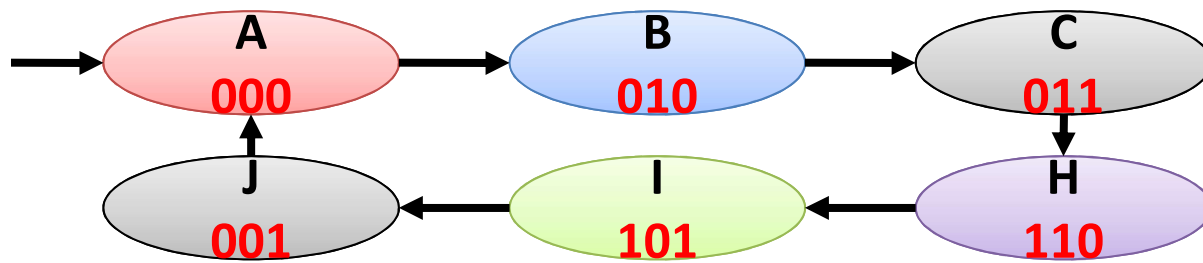
0,2,3,4,5,1,0...



Counter design using JK FF



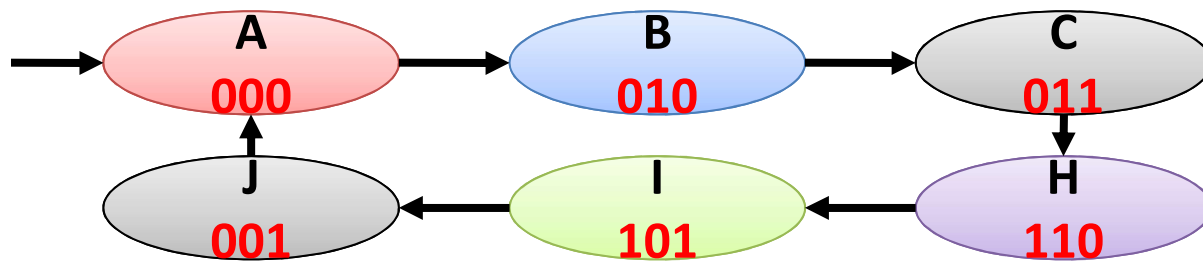
Excitation Table: Sync Counter Using D FF



Q	Q ⁺	D
0	0	0
0	1	1
1	0	0
1	1	1

Present State			Next State			Flip Flop Inputs		
Q1	Q2	Q3	Q1 ⁺	Q2 ⁺	Q3 ⁺	D1	D2	D3
0	0	0	0	1	0	0	1	0
0	1	0	0	1	1	0	1	1
0	1	1	1	1	0	1	1	0
1	1	0	1	0	1	1	0	1
1	0	1	0	0	1	0	0	1
0	0	1	0	0	0	0	0	0

Excitation Table: Sync Counter Using D FF



Q	Q ⁺	D
0	0	0
0	1	1
1	0	0
1	1	1

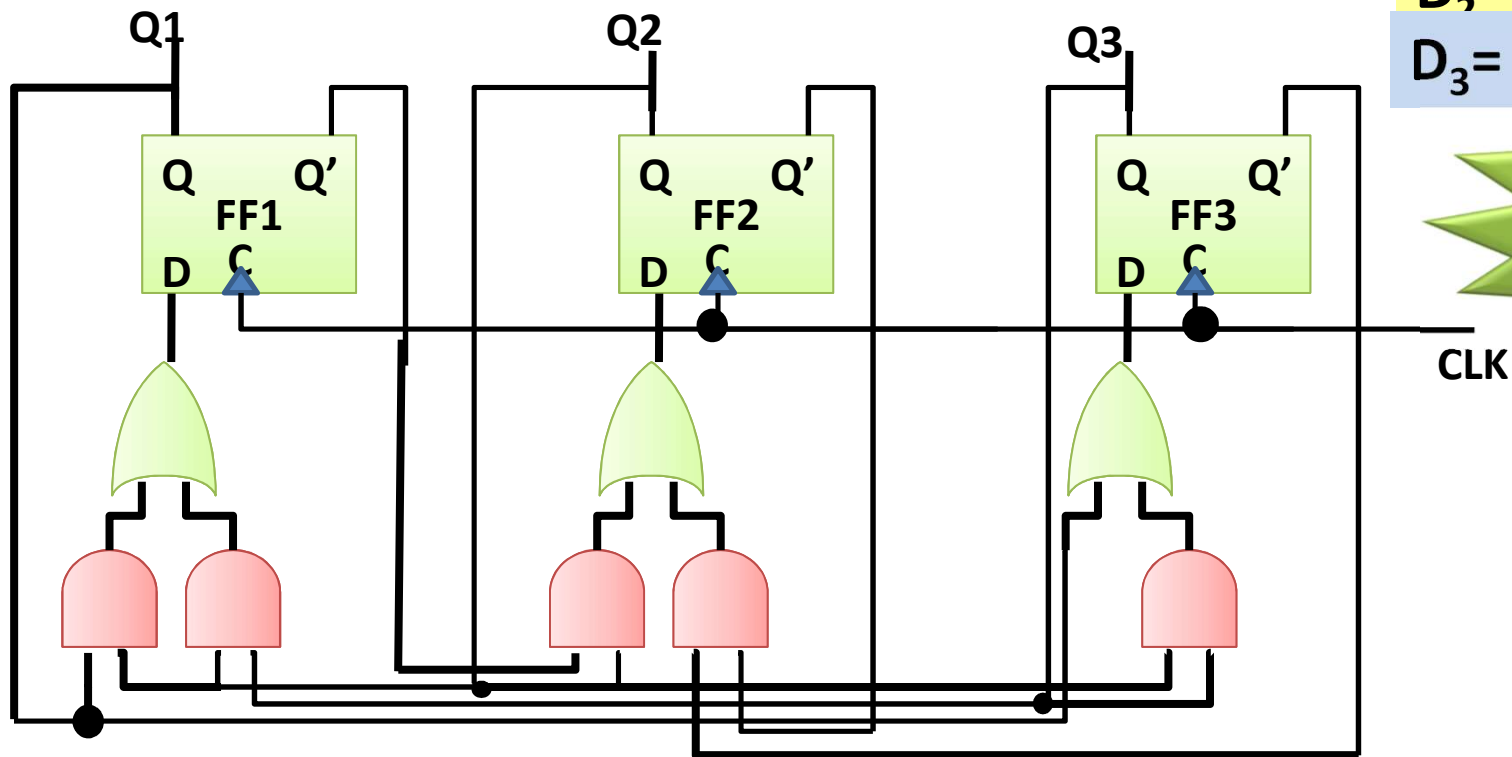
Present State			Flip Flop Inputs		
Q1	Q2	Q3	D1	D2	D3
0	0	0	0	1	0
0	1	0	0	1	1
0	1	1	1	1	0
1	1	0	1	0	1
1	0	1	0	0	1
0	0	1	0	0	0

$$D_1 = Q_1 Q_2 + Q_2 Q_3$$

$$D_2 = Q_1' Q_2 + Q_2' Q_3'$$

$$D_3 = Q_1 + Q_2 Q_3$$

Counter Implementation using D FF



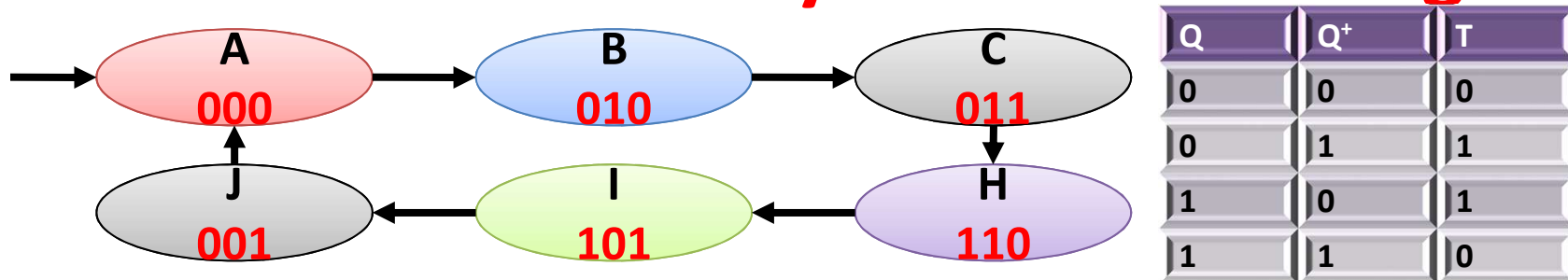
$$D_1 = Q_1Q_2 + Q_2Q_3$$

$$D_2 = Q_1'Q_2 + Q_2'Q_3'$$

$$D_3 = Q_1 + Q_2Q_3$$

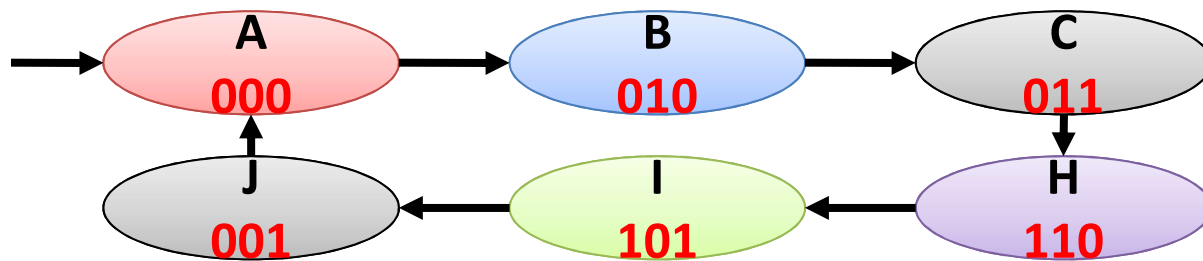
Same as
FSM
Controller

Excitation Table: Sync Counter Using T FF



Present State			Next State			Flip Flop Inputs		
Q1	Q2	Q3	Q1 ⁺	Q2 ⁺	Q3 ⁺	T1	T2	T3
0	0	0	0	1	0	0	1	0
0	1	0	0	1	1	0	0	1
0	1	1	1	1	0	1	0	1
1	1	0	1	0	1	0	1	1
1	0	1	0	0	1	1	0	0
0	0	1	0	0	0	0	0	1

Excitation Table: Sync Counter Using D FF



Q	Q ⁺	T
0	0	0
0	1	1
1	0	0
1	1	1

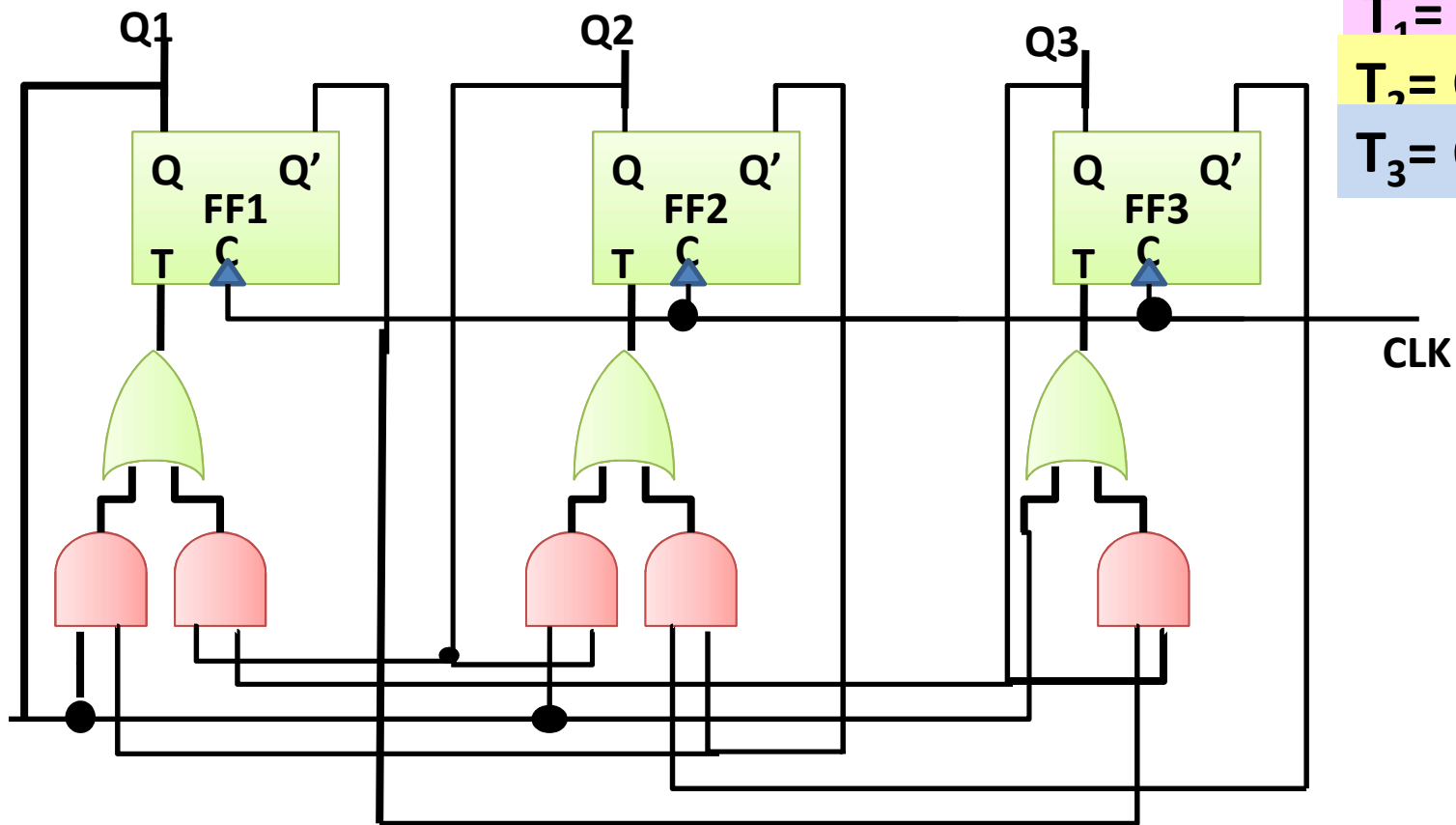
Present State			Flip Flop Inputs		
Q1	Q2	Q3	T1	T2	T3
0	0	0	0	1	0
0	1	0	0	0	1
0	1	1	1	0	1
1	1	0	0	1	1
1	0	1	1	0	0
0	0	1	0	0	1

$$T_1 = Q_1 Q'_2 + Q_2 Q_3$$

$$T_2 = Q_1 Q_2 + Q'_2 Q'_3$$

$$T_3 = Q_1 + Q'_1 Q_3$$

Counter Implementation using T FF

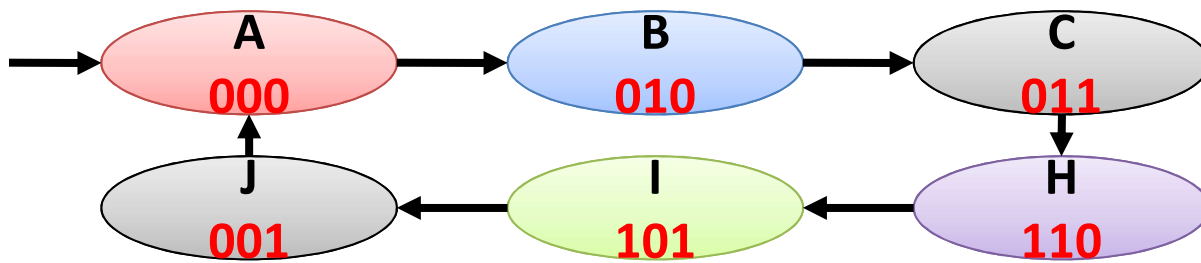


$$T_1 = Q_1 Q'_2 + Q_2 Q_3$$

$$T_2 = Q_1 Q_2 + Q_2' Q_3'$$

$$T_3 = Q_1 + Q_1' Q_3$$

Excitation Table: Sync Counter Using RS FF



Q	Q ⁺	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Present State			Next State			Flip Flop Inputs					
Q1	Q2	Q3	Q1 ⁺	Q2 ⁺	Q3 ⁺	S1	R1	S2	R2	S3	R3
0	0	0	0	1	0	0	X	1	0	0	X
0	1	0	0	1	1	0	X	X	0	1	0
0	1	1	1	1	0	1	0	X	0	0	1
1	1	0	1	0	1	X	0	0	1	1	0
1	0	1	0	0	1	0	1	0	X	X	0
0	0	1	0	0	0	0	X	0	X	0	1

FF Input Functions

Present State			Flip Flop Inputs					
Q1	Q2	Q3	S1	R1	S2	R2	S3	R3
0	0	0	0	X	1	0	0	X
0	1	0	0	X	X	0	1	0
0	1	1	1	0	X	0	0	1
1	1	0	X	0	0	1	1	0
1	0	1	0	1	0	X	X	0
0	0	1	0	X	0	X	0	1

$$S1 = F(Q1, Q2, Q3)$$

$$R1 = F(Q1, Q2, Q3)$$

$$S2 = F(Q1, Q2, Q3)$$

$$R2 = F(Q1, Q2, Q3)$$

$$S3 = F(Q1, Q2, Q3)$$

$$R3 = F(Q1, Q2, Q3)$$

Solve Each Function Using KMAP

Present State			Flip Flop Inputs					
Q_1	Q_2	Q_3	R_1	S_1	S_2	R_2	S_3	R_3
0	0	0	0	X	1	X	0	X
0	1	0	0	X	X	0	1	X
0	1	1	1	X	X	0	X	1
1	1	0	X	0	X	1	1	X
1	0	1	X	1	0	X	X	0
0	0	1	0	X	0	X	X	1

$\xrightarrow{Q_2'Q_3'}$
 $\downarrow Q_1'$

Q_1'	0	0	1	0
Q_1	X	X	X	X

$$S_1 = Q_2 Q_3$$

$$R_1 = Q_2'$$

$$S_2 = Q_1' Q_3'$$

$$R_2 = Q_1$$

$$S_3 = Q_2 Q_3'$$

$$R_3 = Q_1' Q_3$$

Counter Implementation using SR FF

$$S_1 = Q_2 Q_3$$

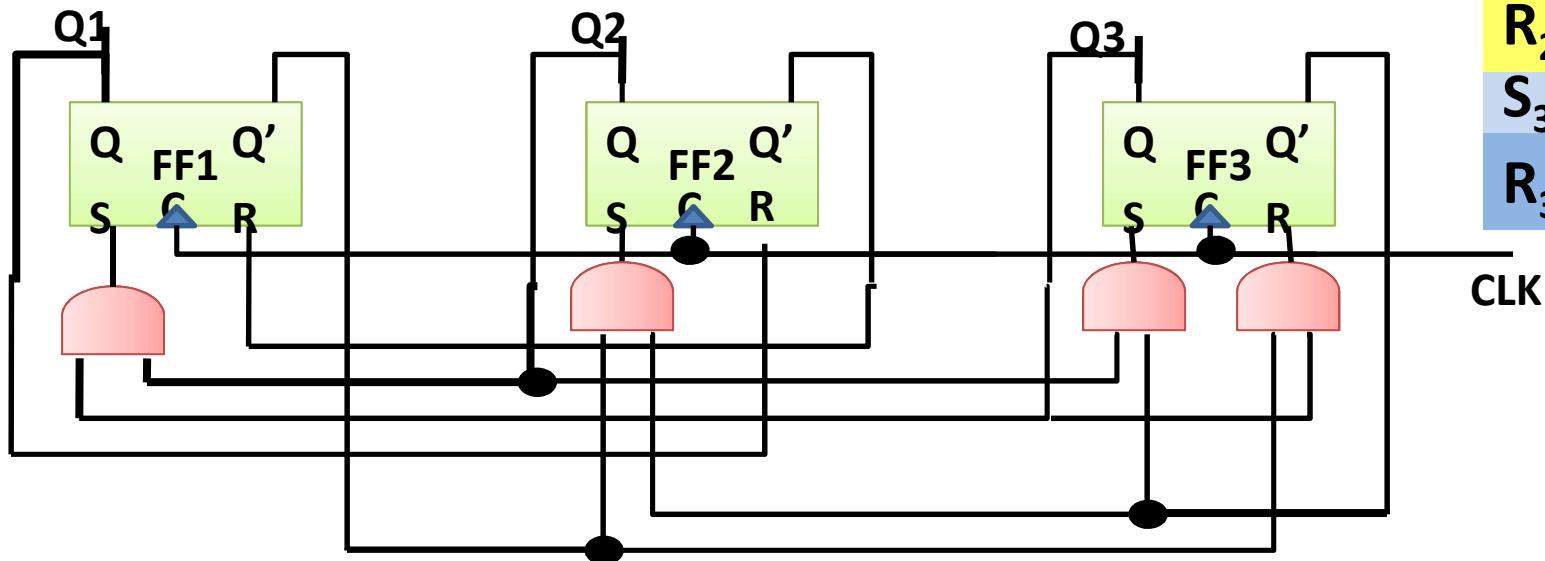
$$R_1 = Q_2'$$

$$S_2 = Q_1' Q_3'$$

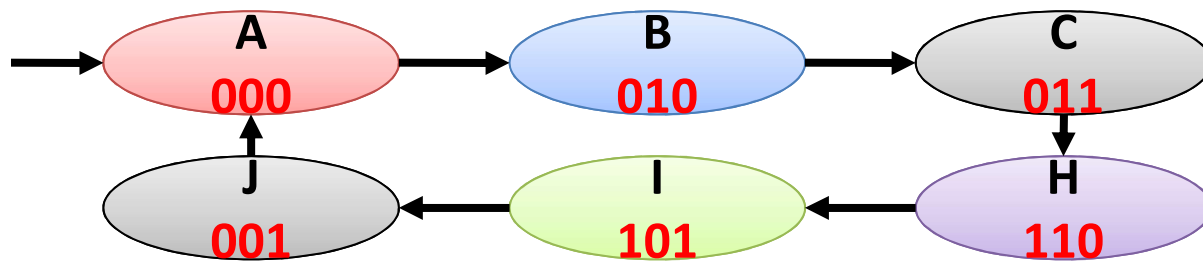
$$R_2 = Q_1$$

$$S_3 = Q_2 Q_3'$$

$$R_3 = Q_1' Q_3$$



Excitation Table: Sync Counter Using JK FF



Q	Q ⁺	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Present State			Next State			Flip Flop Inputs					
Q1	Q2	Q3	Q1 ⁺	Q2 ⁺	Q3 ⁺	J1	K1	J2	K2	J3	K3
0	0	0	0	1	0	0	X	1	X	0	X
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	1	0	1	X	X	0	X	1
1	1	0	1	0	1	X	0	X	1	1	X
1	0	1	0	0	1	X	1	0	X	X	0
0	0	1	0	0	0	0	X	0	X	X	1

FF Input Functions

Present State			Flip Flop Inputs					
Q1	Q2	Q3	J1	K1	J2	K2	J3	K3
0	0	0	0	X	1	X	0	X
0	1	0	0	X	X	0	1	X
0	1	1	1	X	X	0	X	1
1	1	0	X	0	X	1	1	X
1	0	1	X	1	0	X	X	0
0	0	1	0	X	0	X	X	1

$$J1 = F(Q1, Q2, Q3)$$

$$K1 = F(Q1, Q2, Q3)$$

$$J2 = F(Q1, Q2, Q3)$$

$$K2 = F(Q1, Q2, Q3)$$

$$J3 = F(Q1, Q2, Q3)$$

$$K3 = F(Q1, Q2, Q3)$$

Solve Each Function Using KMAP

Present State			Flip Flop Inputs					
Q1	Q2	Q3	J1	K1	J2	K2	J3	K3
0	0	0	0	X	1	X	0	X
0	1	0	0	X	X	0	1	X
0	1	1	1	X	X	0	X	1
1	1	0	X	0	X	1	1	X
1	0	1	X	1	0	X	X	0
0	0	1	0	X	0	X	X	1

$\downarrow$ Q1' $\xrightarrow{\quad}$ Q2'Q3'

0	0	1	0
X	X	X	X

$$J1 = Q2Q3$$

$$K1 = Q2'$$

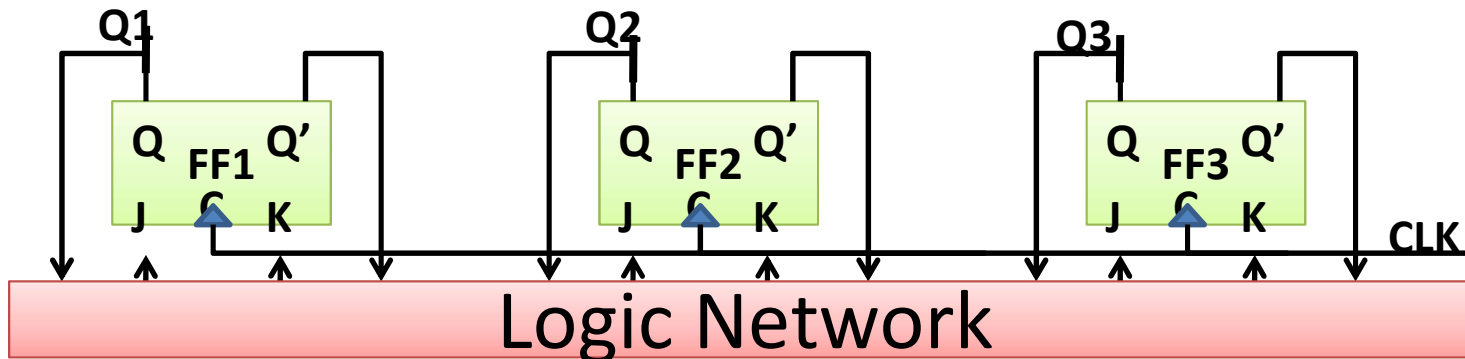
$$J2 = Q3'$$

$$K2 = Q1$$

$$J3 = Q2$$

$$K3 = Q1'$$

Counter Logic Diagram: using JK FF



$$J1 = Q2Q3$$

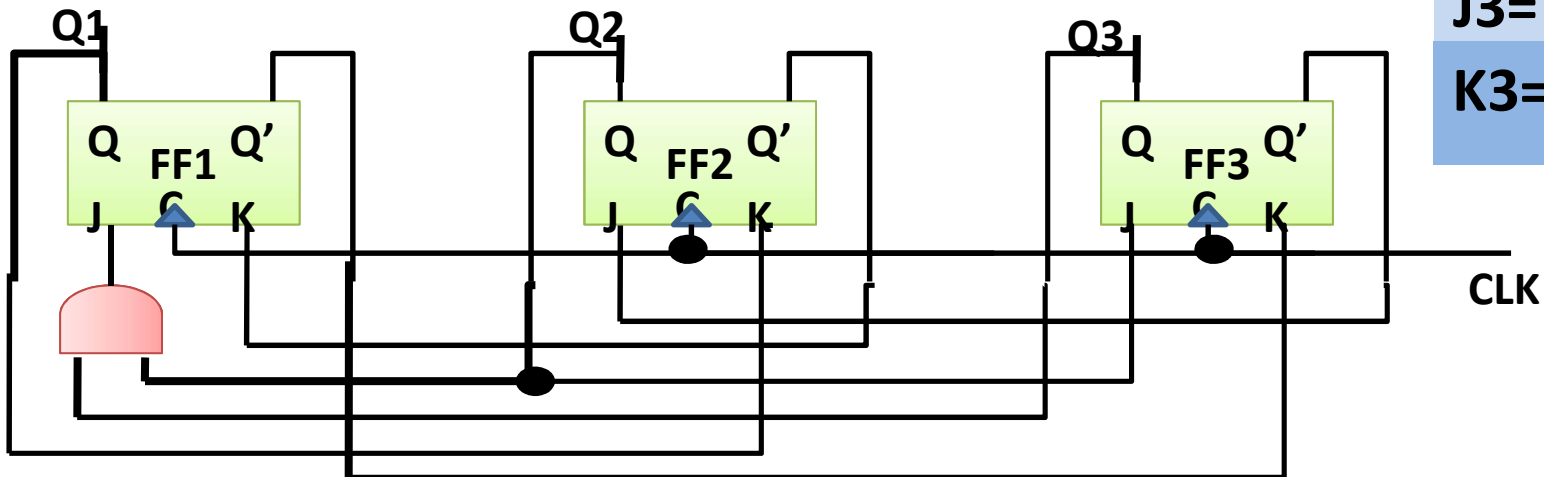
$$K1 = Q2'$$

$$J2 = Q3'$$

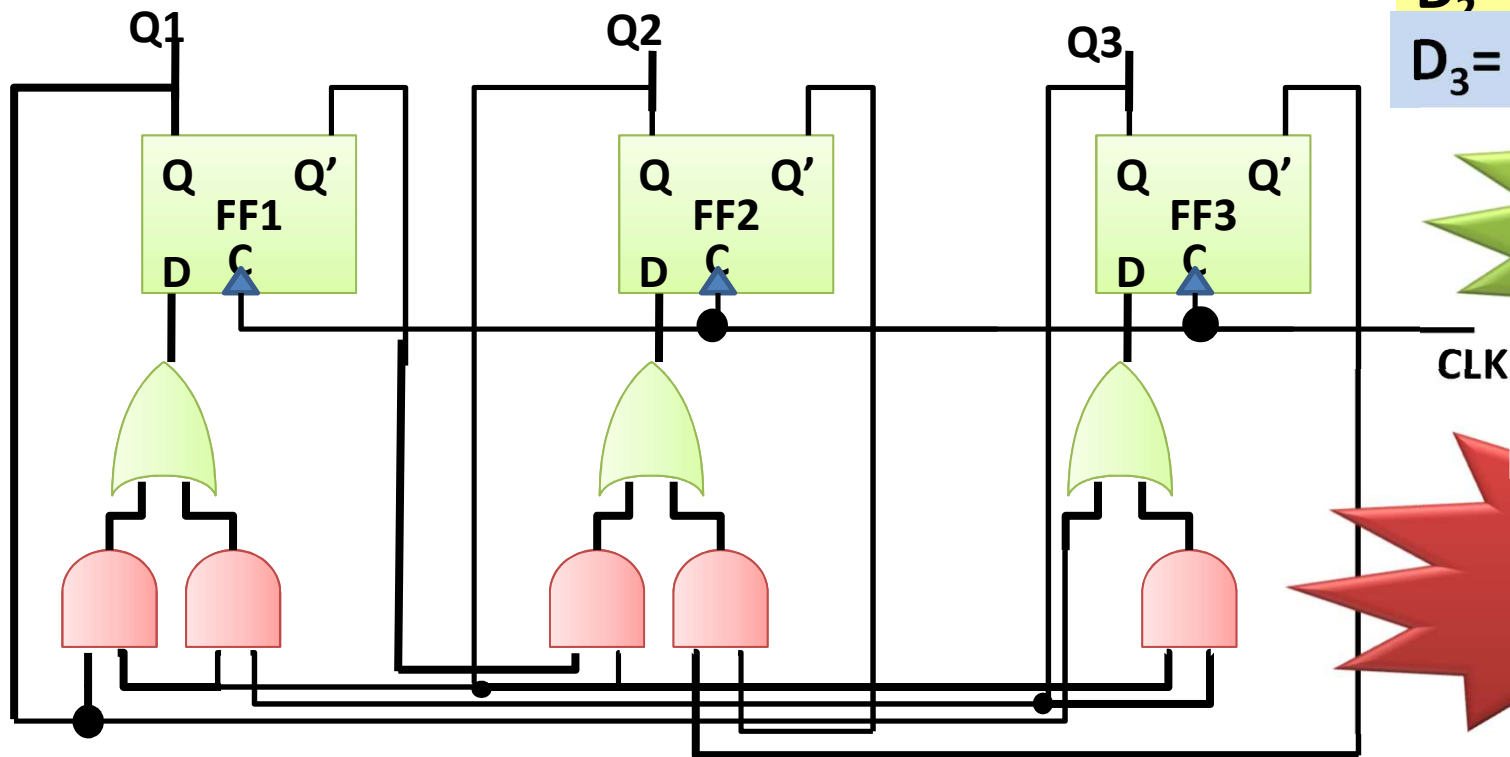
$$K2 = Q1$$

$$J3 = Q2$$

$$K3 = Q1'$$



Counter Implementation using D FF



$$D_1 = Q_1 Q_2 + Q_2 Q_3$$

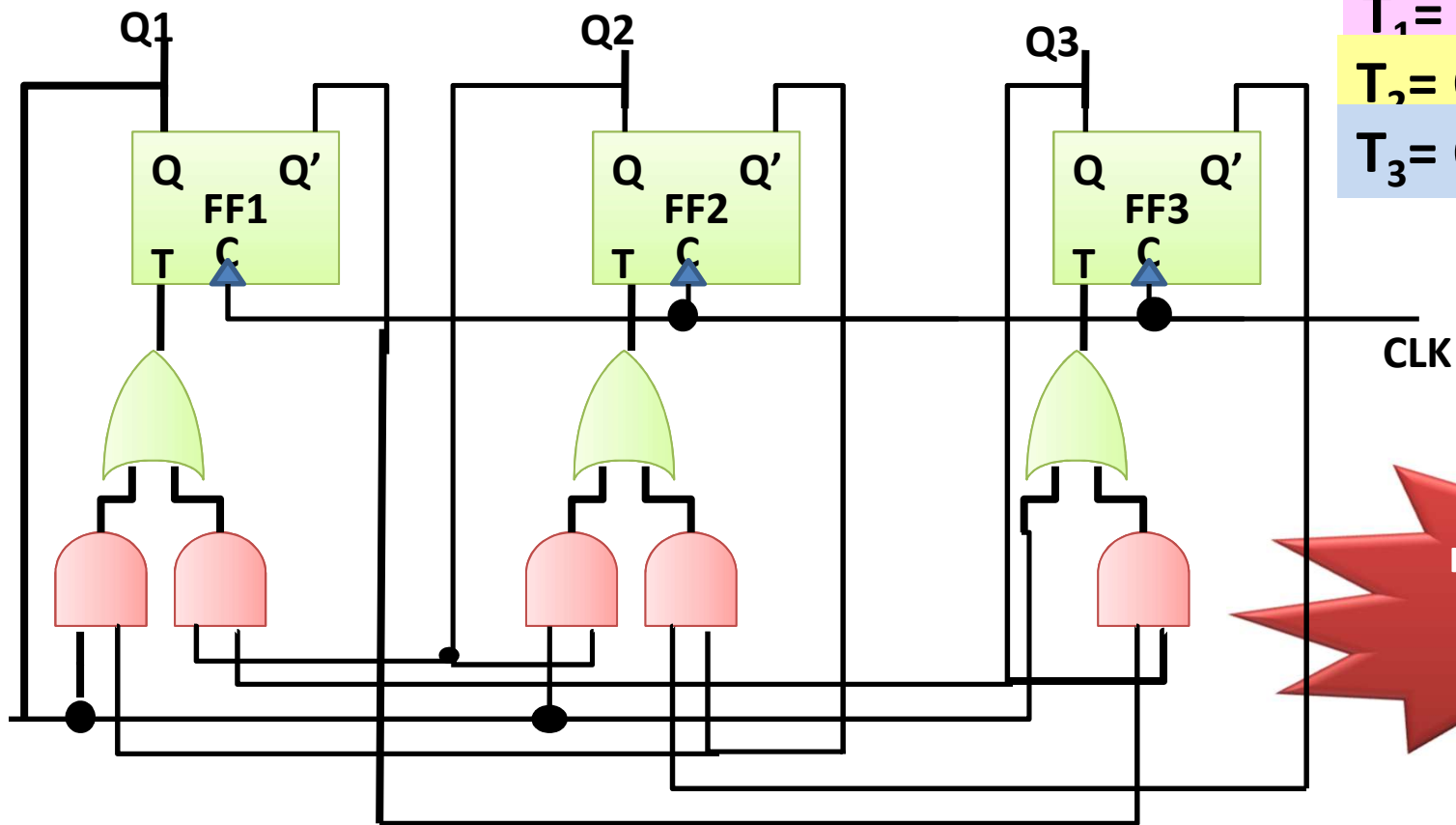
$$D_2 = Q_1' Q_2 + Q_2' Q_3'$$

$$D_3 = Q_1 + Q_2 Q_3$$

Same as
FSM
Controller

Require more
Logic as
compared to
JK FF Based

Counter Implementation using T FF



$$T_1 = Q_1 Q'_2 + Q_2 Q_3$$

$$T_2 = Q_1 Q_2 + Q_2' Q_3'$$

$$T_3 = Q_1 + Q_1' Q_3$$

Still Require
more Logic as
compared to
JK FF Based

Counter Implementation using RS FF

$$S_1 = Q_2 Q_3$$

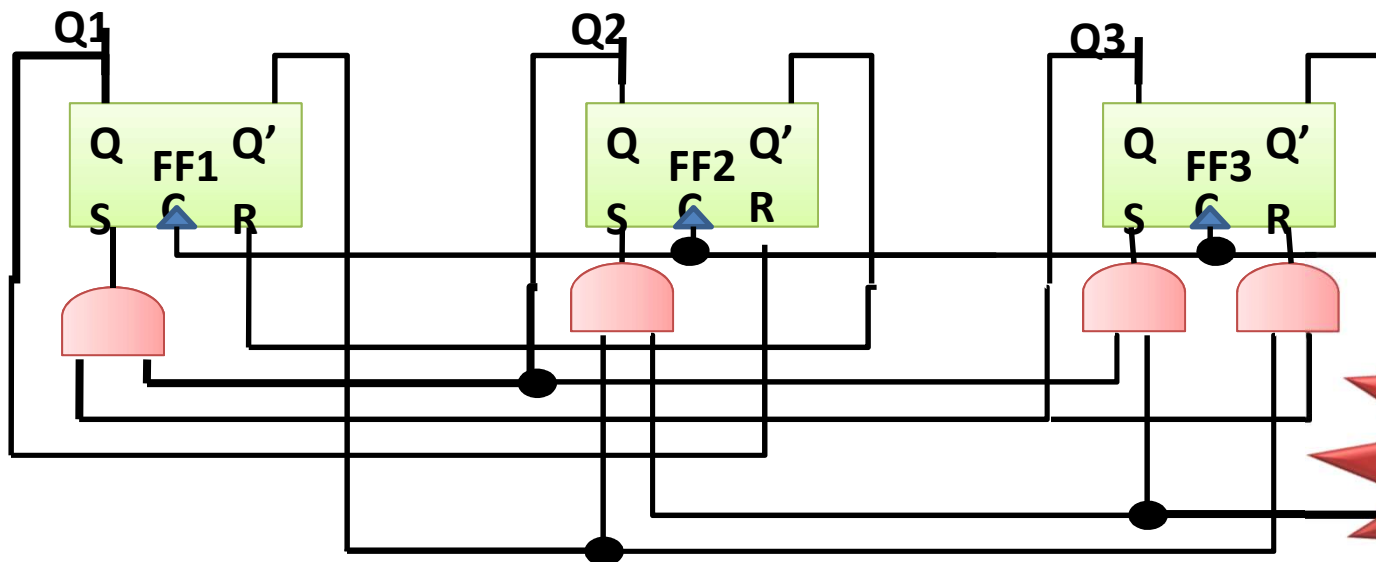
$$R_1 = Q_2'$$

$$S_2 = Q_1' Q_3'$$

$$R_2 = Q_1$$

$$S_3 = Q_2 Q_3'$$

$$R_3 = Q_1' Q_3$$



CLK

Require
more Logic
as JK FF
Based

JK based CCC $\leq$ D based CCC : Why?

- JK FF based Combinational Circuit Complexity (CCC) $\leq$ D FF Based CCC

Q	Q ⁺	D
0	0	0
0	1	1
1	0	0
1	1	1

Q	Q ⁺	T
0	0	0
0	1	1
1	0	1
1	1	0

Q	Q ⁺	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

Q	Q ⁺	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Possibly: Many Don't case get combined very nicely to get optimized circuit

Thanks